Shape sensitivities in JAX-DSMC

From the Boltzmann problem and a windowed objective to the gradient estimator, checked term by term against the code

main, 2026-10-02

Outline

1. The problem: Boltzmann equation on a body-dependent domain, windowed drag objective
2. DSMC as a Markov chain, and every place the geometry enters it
3. The value function (adjoint) and how the code carries it on particles
4. The first-variation identity: pathwise, own-reward, score, boundary-band and operator terms
5. Each term as implemented (`jax_dsmc/shape.py`, `jax_dsmc/ensemble_device.py`, `examples/ensemble_mesh.py`)
6. Audit: what is in, what is deliberately left out, what is missing

Notation follows `docs/math/ensemble_adjoint.tex` ("the note"); equation labels there are cited by name (note eq. `label`).

The kinetic problem

Flow domain $\Omega(\theta) = \text{box} \setminus B(\theta)$, the body given by a C^1 level set

$$B(\theta) = \{x : \phi(x; c(\theta)) < 0\}, \qquad \phi(x; c) = \sum_j c_j B_j(x) \quad \text{(cubic B-splines, spacing } h_\phi),$$

with $n = \nabla\phi/|\nabla\phi|$ the normal into the flow. Boltzmann equation for the one-particle density $f(t, x, v)$:

$$\partial_t f + v \cdot \nabla_x f = Q(f, f) \quad \text{in } \Omega(\theta),$$

with: inflow (drifting Maxwellian flux) at $x = 0$; outflow at $x = L$; specular y-walls; and on the body, for $v \cdot n > 0$,

$$|v \cdot n|\, f(x, v) = \int_{v' \cdot n < 0} K(v' \to v;\, n)\, |v' \cdot n|\, f(x, v')\, \mathrm{d}v', \qquad K = (1 - a)\, \delta(v - R_n v') + a\, q_w(v \mid n),$$

$R_n v' = v' - 2(v' \cdot n)n$, q_w the diffuse half-Maxwellian at the wall temperature (note eq. `diffuse-density`). The squash study and `optc` use specular bodies ($a = 0$).

Controls: `radius`, `centre_x`, `mode2`, `width`, `squash_y` (chained through $c(\theta)$), or the coefficients c themselves (the optimiser).

The windowed objective

Mean x-force on the body over the window $[t_1, t_2]$, $t_2 - t_1 = W\Delta t$:

$$D(\theta) = \frac{1}{W\Delta t} \int_{t_1}^{t_2} \int_{\partial B} \int_{v\cdot n<0} m \left(v - v^{\mathrm{out}}(v)\right)_x |v \cdot n| \, f \mathrm{d}v \mathrm{d}S \mathrm{d}t$$

(for a diffuse wall, v^{out} is averaged over K). In DSMC, with F_{num} real molecules per simulator particle, this becomes an expectation of a sum of *per-hit rewards*:

$$D = \mathbb{E}\Big[\sum_{k \in \text{window}} \; \sum_{\text{body hits } j \text{ in step } k} r_{k,j} \Big], \qquad r_{k,j} = \frac{F_{\mathrm{num}}}{W\Delta t} \, m \, (v_{k,j}^{\mathrm{in}} - v_{k,j}^{\mathrm{out}})_x .$$

Code: `ensemble_mesh._drag_reward` (per hit, zero outside the window and off the body), `objective_value` (the same sum from the surface tallies; this is what gets reported).

Squash study (`examples/mesh/optimize_squash.py`): $J(s) = D(s) - \lambda_A \, A(s)$, with $A(s) = \pi R^2 s$ exact; λ_A is fixed so that the free-molecular optimum is at $s = 0.35$. Only $\mathrm{d}D/\mathrm{d}s$ needs the adjoint.

There is no terminal or per-step state reward: `phi_final` $= 0$, `phi_step` unused.

DSMC as a Markov chain (`simulate._step`)

State $Z_k = \{(x_i, v_i, a_i)\}_{i=1}^{N}$ (fixed slots, alive mask). One step, in order:

$$Z_k \xrightarrow{\text{move + walls } \mathcal{W}_\theta} Z_k^m \xrightarrow{\text{inject}} \cdot \xrightarrow{\text{sort}} \cdot \xrightarrow{\text{collide (NTC/VSS) } \mathcal{C}_\theta} Z_{k+1}.$$

RNG contract: every stage draws from $\mathrm{fold_in}(\mathrm{fold_in}(\kappa, k), \text{stage})$, so with the key fixed each step is a deterministic function of its input. The record (`EventRecord`) stores Z_k^m, the collision pairs, births, and wall events.

Where θ enters the forward:

(G1) the move: hit times t^h, hit points x^h, normals n, specular $v^{\mathrm{out}} = R_n v^{\mathrm{in}}$, and multi-bounce chains;

(G2) the per-hit reward itself, through $v^{\mathrm{out}}(n)$;

(G3) the diffuse kernel $q_w(\cdot \mid n)$ (its normal);

(G4) which particles hit (hit set), and in which step;

(G5) NTC: the attempt count $\bar{n}_{\mathrm{att},c} = \frac{1}{2} N_c (N_c - 1) \Sigma_c \, \Delta t \, F_{\mathrm{num}} / V_c$ uses the *cut-cell flow volume* $V_c(\theta)$;

(G6) inject: entrants are tracked for a random fraction of Δt (birth-step hits).

The value function

For the expected objective $D = \mathbb{E}[\sum_k R_k]$ define the value of a particle state at the post-move instant of step k:

$$\lambda_k(x, v) = \frac{\delta D}{\delta f_k^m(x, v)} \;\approx\; \mathbb{E}\Big[\text{rewards collected from } t_k^m \text{ on } \Big|\ \text{one particle at } (x, v)\Big],$$

one simulator particle standing for F_{num} molecules. For a mean-field (Boltzmann) gas, λ solves the adjoint equation

$$-\partial_t \lambda - v \cdot \nabla_x \lambda = \mathcal{L}_f^* \lambda + (\text{reward sources}),$$

$$(\mathcal{L}_f^* \psi)(v) = \int f(w) \int q\,[\psi(v') + \psi(w') - \psi(v) - \psi(w)]\,\mathrm{d}\sigma\,\mathrm{d}w,$$

with, on a specular body, $\lambda(x, v^{\mathrm{in}}) = r(v^{\mathrm{in}}) + \lambda(x, R_n v^{\mathrm{in}})$ (the hit pays its reward and the particle continues).

The code never solves this PDE. It carries **one value per particle slot, at that slot's own state**, going backward through the record (route (b) of the review; note §3).

The backward recursion as implemented

Backward through step k, with μ the values at t_{k+1}:

1. **Collision pull-back** $\mu \mapsto \lambda_k^m$ (`_substep`, `_partner_pass`): S substeps, K virtual partners j_κ per particle, all four values from one per-cell fit Λ:

$$\lambda_i \mathrel{+}= \sum_\kappa w_{i\kappa}\left[\Lambda(v_i') + \Lambda(v_{j_\kappa}') - \Lambda(v_i) - \Lambda(v_{j_\kappa})\right], \qquad w_{i\kappa} = \frac{q(g_{i\kappa})}{\Sigma_c}\,\frac{2\,n_{\mathrm{att},c}}{N_c}\,\frac{1}{KS}$$

   (note eqs. `virtual`, `virtual-weight`); staged nodes move values across substep boundaries (note eq. `boundary-move`). $(S, K) = (4, 8)$ collisional, $(1, 2)$ collisionless.

2. **Move pull-back** (`_move_pullback`): survivors carry their value back, plus the rewards of the hits they took during the move:

$$\mu_i^{(k)} = \mathbf{1}[a_{k,i}^m \wedge \neg \mathrm{born}_{k,i}]\Big(\lambda_{k,i}^m + \sum_j r_{k,j,i}\Big).$$

   Newborns and particles that leave the box during the move get 0.

The output is λ_k^m at every post-move state (`Backward.lam`, $M \times N$). The geometry never enters this sweep, except through the record it reads.

The per-cell field interpolant (`ensemble_device.field_query`)

Nodes: alive particles of the query's cell (or the 3^D block of cells when pooled), in scaled coordinates $\tilde{p} = (x/\Delta x,\ v/v_{\mathrm{scale}})$, $v_{\mathrm{scale}} = \sqrt{kT/m}$. The k nearest nodes ($k = 32$; 96 pooled) get weights $\omega_n = 1/(1 + (d_n/d_k)^2)$.

Design ("quadratic_full", $1 + D + 3 + 6 + D$ columns, 14 in 2-D):

$$\phi(\delta) = \big(1,\ \delta_x,\ \delta_v,\ \{\delta_{v,a}\delta_{v,b}\}_{a\le b},\ \delta_x^2\big), \qquad \Big(\sum_n \omega_n \phi_n \phi_n^\top + \rho\bar{d}\,\mathrm{diag}(0,1,\ldots,1)\Big)\beta = \sum_n \omega_n \phi_n \lambda_n,$$

solved by an unrolled Cholesky factorisation (`_spd_solve`; both value columns at once since 2026-10-02).

$$\Lambda = \beta_0, \qquad \nabla_x \Lambda = \beta_{1:1+D}/\Delta x, \qquad \nabla_v \Lambda = \beta_{1+D:4+D}/v_{\mathrm{scale}}.$$

The collision pull-back uses the value; the shape gradient uses the two slopes (`node_gradients`).

The first-variation identity for one step

Write one step as $Z' = G_\theta(Z, \omega)$ with randomness $\omega \sim p_\theta$, reward $R(Z, Z')$ and value $V(Z')$ of what follows. Then

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\mathbb{E}\big[R + V(Z')\big] = \underbrace{\mathbb{E}\big[(\partial_{Z'} R + \nabla V)\cdot \partial_\theta G\big]}_{\text{pathwise (frozen } \omega)} + \underbrace{\mathbb{E}\big[\partial_\theta R\big|_{Z'}\big]}_{\text{own reward}}$$
$$+ \underbrace{\mathbb{E}\big[(R+V)\,\partial_\theta \log p_\theta(\omega)\big]}_{\text{score}} + \underbrace{\int_{\Sigma_\theta} [\![R+V]\!]\, u_\Sigma \,\mathrm{d}\sigma}_{\text{boundary band}}$$

where Σ_θ is the set of ω at which G_θ jumps (a hit becomes a miss, or a hit moves from one step into the next), $[\![\cdot]\!]$ the jump across it, and u_Σ the speed at which Σ_θ moves with θ. Summed over steps, with $V = \lambda_k$:

$$\frac{\mathrm{d}D}{\mathrm{d}\theta} = \sum_k \Big(\text{pathwise}_k + \text{own}_k + \text{score}_k + \text{band}_k\Big) + \underbrace{\sum_k \mathbb{E}\big[\partial_\theta(\text{collision operator}) \text{ at fixed particles}\big]}_{\text{operator term (G5)}}.$$

Each term below is matched to one of these pieces. None of them differentiates the estimator of λ: the value is a cotangent, not a function of θ.

The replayed wall map (`shape.replay`, `shape.wall_map`)

Under the RNG contract the move of step k is a pure function

$$(x_k^{\mathrm{post}}, v_k^{\mathrm{post}}, \log p_{\varepsilon,k}, T_k, \mathcal{E}_k) = \mathcal{W}_\theta(x_k, v_k, \kappa_k)$$

(T_k surface tallies, $\mathcal{E}_k$ traced events: $x^h, v^{\mathrm{in}}, v^{\mathrm{out}}, n, t^h$ per hit).

- `shape.replay` re-runs $\mathcal{W}_\theta$ from the record, step by step (`lax.scan`), forms the per-hit rewards and timing rewards, and checks the replay: any flipped outcome is an error; continuous error $\leq 10^{-9}$ (`REPLAY_TOL`).
- `shape._gradient_step` takes `jax.vjp` of $\mathcal{W}_\theta$ w.r.t. the geometry pytree (level-set coefficients) and pulls back *all* cotangents of the step in one batched transpose (`_pullbacks`).

Only the survivors of the step contribute: $a_k \wedge a_k^m \wedge \neg\mathrm{born}_k$. θ derivatives inside $\mathcal{W}$: the hit time solves $\phi(x+tv) = 0$, so $\partial t^h/\partial c_j = -B_j(x^h)/(v \cdot \nabla\phi(x^h))$ (bisection, then one Newton step that carries the derivative), then $x^h = x + t^h v$, $n(x^h; c)$, $v^{\mathrm{out}} = R_n v$, $x^{\mathrm{post}} = x^h + v^{\mathrm{out}}(\Delta t - t^h)$, repeated for each bounce. Grazing hits: the $1/(v \cdot \nabla\phi)$ is floored at $\epsilon|v||\nabla\phi|$ in the tangent only (`levelset.regularised_quotient`; $\epsilon =$ `grazing`, bias $O(\epsilon)$).

Term 1 — pathwise x, v (G1)

$$\text{pathwise}_k = \sum_{i \in \text{surv}} \nabla_x \lambda_k^m(x_i^{\text{post}}, v_i^{\text{post}}) \cdot \partial_\theta x_i^{\text{post}} + \nabla_v \lambda_k^m(x_i^{\text{post}}, v_i^{\text{post}}) \cdot \partial_\theta v_i^{\text{post}}.$$

- Cotangents from `node_gradients` at the record's post-move state; rows `pathwise_x`, `pathwise_v` of `ShapeTerms`.
- `hit_fits=True`: fits only at slots with a body event in the step. With $\varepsilon = 0$, every other slot has $\partial_\theta x^{\text{post}} = \partial_\theta v^{\text{post}} = 0$ exactly, so the result is the same to the last bit at about $1/200$ of the fitting work.
- **One-sided**: nodes restricted to $v \cdot n > 0$ for the query's last hit normal. A particle just reflected is fitted from particles that also just left the wall, not from the ones about to hit it.
- **Hit-ahead class** (`shape.hit_ahead`): nodes also restricted to the query's class $\kappa =$ "this particle hits the body at a later captured step" (note eq. `hit-ahead`).

Term 1, $\nabla_x\lambda$: the along-flight identity

Collisionless: the state $(x - sv, v)$ is the same trajectory delayed by s. A delay changes the objective only by moving hits across step boundaries, at rate $T =$ the value of the *timing rewards* τ (Term 4):

$$\lambda(x - sv, v) = \lambda(x, v) + s\,T(x, v) + O(s^2) \quad \Longrightarrow \quad v \cdot \nabla_x\lambda = -T.$$

T is carried by a second backward (`hit_reward` $= \tau$); since 2026-10-02 both run as one two-column sweep (PSEUDOCODE 9.7.1). `shape.along_flight` replaces the component of $\nabla_x\lambda$ along v and keeps the fitted perpendicular part:

$$\nabla_x\lambda \leftarrow \nabla_x\lambda - \frac{v \cdot \nabla_x\lambda + T}{|v|^2}\, v.$$

With collisions the backward equation $\partial_t\lambda + v \cdot \nabla_x\lambda + \mathcal{L}_f^*\lambda + r = 0$ gives $v \cdot \nabla_x\lambda = -T - \mathcal{L}_f^*\lambda$: the extra flight time is spent colliding at the rate of the state. Since 2026-10-02 the code adds it: the virtual pull-back returns each slot's increment $\Delta t\,\mathcal{L}^*\lambda = \sum_{\text{substeps}} \sum_{\kappa} w\Phi$ (`Backward.collision`), and `gradient_variants(collision=)` uses $T + \mathcal{L}^*\lambda$ per step.

Default estimator: `along_pool` (pooled 3^D block, $k = 96$, plus along-flight), used by `optimize.py` and `optimize_squash.py`.

Term 2 — the reward's own dependence: the tally term (G2)

The drag reward $r = \frac{F_{\mathrm{num}}}{W\Delta t}\, m(v^{\mathrm{in}} - v^{\mathrm{out}})_x$ depends on θ through $v^{\mathrm{out}} = R_n v^{\mathrm{in}}$ (and, for multi-bounce chains, through v^{in} of later bounces). The surface tallies are outputs of $\mathcal{W}_\theta$, so the term is one more cotangent:

$$\mathrm{own}_k = \frac{\partial D}{\partial T_k} \cdot \partial_\theta T_k, \qquad \frac{\partial D}{\partial T_k} = \frac{F_{\mathrm{num}}}{W\Delta t}\, \mathbf{1}[k \in \text{window}] \text{ on the body rows' } x\text{-momentum.}$$

Code: `tally_cotangent` × `tally_weight` in `_gradient_step`; row `pathwise_tally`.

On the specular cylinder in free-molecular flow this term alone is $\mathrm{d}F/\mathrm{d}r = F/r$ ($F \propto \int \cos^2\varphi\, \mathrm{d}b = 4r/3$; the new grazing rays carry $\cos^2\varphi = 0$). Measured: 1.27 ± 0.02 vs $F/r = 1.24$ ($\times 10^{-20}$). The $\nabla\lambda$ terms carry the flow-field (blockage) feedback.

Term 3 — the score terms (G3)

Diffuse normal. The re-emitted velocity is `stop_gradient`; its law depends on n:

$$q_w(v \mid n) = \frac{v_n}{s^2} e^{-v_n^2/2s^2} \frac{e^{-|v_t-u|^2/2s^2}}{2\pi s^2}, \qquad \text{score}_k = \sum_j w_{k,j} \nabla_n \log q_w(v_{k,j}^{\text{out}} \mid n_{k,j}) \cdot \partial_\theta n_{k,j}.$$

Weight: the value from the draw onward. With a tally cotangent (the drag runs) the hit's own reward is pathwise through the reparameterised sample, so $w_{k,j} = \lambda^m + \sum_{j'>j} r_{k,j'}$ (`to_go.post`); otherwise $\lambda^m + \sum_{j' \geq j} r_{k,j'}$ (`to_go.pre`). $1/v_n$ is floored at $\epsilon|v^{\text{out}}|$ (bias $\approx \epsilon/2$ of the v_t/v_n part).

Move log-probability (ε-regularised flight time, accommodation draw): cotangent `to_go.start` on $\log p_{\varepsilon,k}$. At $\varepsilon = 0$ and a fixed accommodation it does not depend on the geometry.

For specular bodies (`optc`, squash) the whole score term is exactly zero.

Term 4 — the hit-timing band (G4, step assignment)

A displacement of the surface shifts hit times; a hit near the end of step k crosses into step $k+1$. If the objective changes by $\Delta J_{k,j}$ when it does, the boundary term is (density of hit times at the step end) $\times(-\Delta J)$, estimated over a band $\beta\Delta t$:

$$c_{k,j} = -\frac{\Delta J_{k,j}}{\beta \Delta t}\,\mathbf{1}\big[t^h_{k,j} > (1-\beta)\Delta t\big], \qquad \text{cotangent on } t^h_{k,j} \quad (\beta = 0.2).$$

For the per-hit drag, $\Delta J = +r$ on the window's last step and $-r$ on the step before it (`_drag_jump`): the term lives at the two window edges. On a steady window these cancel in expectation (gate G3a), so `ensemble_mesh` reports it as `timing` and **leaves it out** of `gradient`.

The same $c_{k,j}$ are the *timing rewards* τ whose value T the along-flight identity uses (Term 1).

Term 5 — the silhouette band (G4, hit ↔ miss)

$\phi_{\min} = \min_s \phi(x + s\hat{v})$ along the undeflected line: < 0 hits, > 0 misses. A surface displacement flips the outcome where $|\phi_{\min}| < |\delta\phi|$:

$$\left.\frac{\mathrm{d}D}{\mathrm{d}\theta}\right|_{\mathrm{sil}} = -\mathbb{E}\Big[\rho(0)\,(V_{\mathrm{hit}} - V_{\mathrm{miss}})\,\frac{\partial\phi_{\min}}{\partial\theta}\Big] \approx -\frac{1}{2\varepsilon_b}\sum_{|\phi_{\min,i}|<\varepsilon_b}(V_{\mathrm{hit},i} - V_{\mathrm{miss},i})\,B_j(x_i^*)\,\frac{\partial c_j}{\partial\theta},$$

($\partial\phi_{\min}/\partial c_j = B_j(x^*)$ by the envelope theorem). The realised outcome's value is the reward-to-go; the virtual one is the fitted λ at the virtual post-move state plus the virtual hit's reward (`silhouette_cotangent`). Each passing is counted once per trajectory.

Specular: $V_{\mathrm{hit}} \to V_{\mathrm{miss}}$ at grazing incidence, so the term is $O(\varepsilon_b)$; the runs use `silhouette_band = 0`. Diffuse: $O(1)$ (plate gate $|z| \le 1.6$).

Assembly and the control chain

Per seed (`ensemble_mesh.shape_terms` → `shape.gradient_variants`):

$$\widehat{\nabla_c D} = \underbrace{\texttt{pathwise_x} + \texttt{pathwise_v} + \texttt{pathwise_tally}}_{\texttt{pathwise}} + \texttt{score} + \texttt{silhouette},$$

with `timing` reported but not added. Pipeline: forward with capture → `shape.replay` (rewards r, timing rewards τ, replay check) → two-column value backward (λ from r, T from τ) → `shape.gradient` (VJP per step, all cotangents batched).

Chain to a scalar control: `control_gradient` takes `jax.vjp` of $\theta \mapsto c(\theta)$ (`levelset_coef`); e.g. $\partial c/\partial r = -1$ at every knot for the radius.

Optimiser (`optimize.py`): seed mean g, band mask + Gaussian smoothing (H^1-type), Adam, fixed largest step $s_{\max} = 0.05$, reinitialise ϕ to a signed distance, restore the area by a uniform shift (a projection onto $\oint V_n = 0$). Squash: scalar Adam on $J'(s) = \widehat{\mathrm{d}D/\mathrm{d}s} - \lambda_A \, \mathrm{d}A/\mathrm{d}s$.

Audit: every θ-dependence of the forward

dependence	term	code	status
(G1) hit times, points, specular v^{out}	pathwise x, v	VJP of $\mathcal{W}_\theta$	**in**
(G2) reward through v^{out}	tally	`tally_cotangent`	**in**
(G3) diffuse kernel normal	score	`normal_score`	**in** (0 specular)
(G3) ε flight time, accommodation	score	`to_go.start`	**in** (0 at ε=0)
(G4) hit $\leftrightarrow$ miss	silhouette	`silhouette_cotangent`	**in** (0 if specular)
(G4) hit crosses a step boundary	timing	`timing_cotangent`	**left out** (edges)
(G5) cut-cell volume in NTC	operator	`volume_cotangent`	**in** (2026-10-02)
(G6) entrants' birth-step hits	—	masked out	**left out** (monitored)
hits by particles leaving in the step	—	masked out	**left out** (monitored)
estimator of $\nabla_x \lambda$ along v	along-flight	`along_flight`	**in** ($+\mathcal{L}^* \lambda$, 10-02)
hit-ahead class of the fits	—	`hit_ahead`	**biased**; default now off

The VJP itself is complete for the move: every floating output of $\mathcal{W}_\theta$ gets a cotangent, and the replay is checked against the record at every step.

Finding 1: the hit-ahead class is biased by construction

`shape.hit_ahead` sets $\kappa_{k,i} = \exists\, k' > k$: slot i has a body hit at k'. The fits of $\nabla\lambda$ then use only nodes of the query's own class. This is the library default (`shape.gradient`, `ensemble_mesh.default_config`), so `optimize.py` (the `optc` study) runs with it. The squash and squashkn studies pass `value_class="none"`.

Open since 2026-09-25: the fine $\cos 2\theta$ gate reads -2.386 ± 0.005 against CRN FD -1.59 ($z \approx -120$). The suspected cause was slot refills: the class follows the slot, so a departed particle inherits the hits of the newborn that takes its slot.

That mislabelling is real (coarse circle, seed 1, `scratch/shape_audit/hit_ahead_refill.py`):

- 57 % of slots are refilled within the 300 captured steps;
- 30 % of the hit-ahead labels come from a later occupant;
- 16 % of the fitted queries (`hit_fits`) are mislabelled.

Corrected class: $\kappa_k = h_{k+1} \vee (\kappa_{k+1} \wedge \neg\,\mathrm{born}_{k+1})$, a reverse `lax.scan`. The paired gate on the next slide shows that the corrected class makes the bias worse.

Finding 1: paired gate (`scratch/shape_audit/gate_class_fix.py`)

Coarse circle, level-set body, `mode2` ($\cos 2\theta$), 4 CRN seeds (32–35), the same records for every row; FD $= -1.594 \pm 0.043$, tally term -1.596 ($\times 10^{-20}$), so the $\nabla\lambda$ parts should sum to FD$-$tally $= +0.01 \pm 0.68 \times 10^{-21}$.

fit class	x part 10^{-21}	v part 10^{-21}	gradient 10^{-20}	gradient $-$ FD (paired) 10^{-20}
no class (`none`)	-0.28	$+0.36 \pm 0.15$	-1.587 ± 0.030	$+0.007 \pm 0.071$
hit-ahead, current	-0.72	-6.02 ± 0.34	-2.269 ± 0.036	-0.675 ± 0.057
hit-ahead, stopped at the next birth	-0.80	-10.82 ± 0.62	-2.758 ± 0.069	-1.164 ± 0.091

Refilling is not the cause. A purer class makes the bias *larger*. The stratified fit estimates $\nabla\lambda_\kappa$ within each class, while

$$\lambda = P\lambda_1 + (1-P)\lambda_0 \;\Rightarrow\; \nabla\lambda = P\nabla\lambda_1 + (1-P)\nabla\lambda_0 + \underbrace{\nabla P(\lambda_1 - \lambda_0)}_{\text{dropped}},$$

with $P(z)$ = probability of a later body hit. A reflected particle can come back, most likely after a specular y-wall bounce (not checked), so $\nabla_v P \neq 0$, and $\lambda_1 - \lambda_0 \approx$ the next hit's drag. Neither the silhouette nor the timing term carries this. On the capsule, $P \approx 0$ after a reflection, so the class was harmless there.

Finding 2: the cut-volume (operator) term (built 2026-10-02)

The NTC attempt count carries $1/V_c$, and $V_c(\theta)$ is the cut cell's flow volume. Neither the replay nor the VJP sees the collision step (`with_geometry` rebuilds normals, not volumes). The missing term has the same form as the kernel-parameter term (note eq. `kernel-term`), with $\partial_\theta \log q$ replaced by $\partial_\theta \log V_c^{-1}$:

$$\left.\frac{\mathrm{d}D}{\mathrm{d}\theta}\right|_V = -\sum_k \sum_{c\ \mathrm{cut}} \frac{\partial_\theta V_c}{V_c}\, C_{k,c}, \qquad C_{k,c} = \sum_{i\in c} \sum_\kappa \tfrac{1}{2}\, w_{i\kappa}\, \Phi_{i\kappa},$$

$C_{k,c}$ being the expected change of D from the collisions of cell c in step k, and

$$\frac{\partial V_c}{\partial c_j} = \int_c \delta(\phi)\, B_j \,\mathrm{d}x = \int_{\Gamma\cap c} \frac{B_j}{|\nabla\phi|}\,\mathrm{d}s,$$

with the same for each sub-cell of a split cell.

- Collisionless (Kn $\sim 10^{18}$: radius, `mode2`, `optc`, squash): $C = 0$, so the term vanishes.
- **squashkn (Kn $= 0.5$, M 0.32 and M 10)**: nonzero; the cluster runs omitted it.
- **Built 2026-10-02**: $C_{k,c} = \frac{1}{2}\sum_{i\in c} \Delta t\,(\mathcal{L}^*\lambda)_i$ accumulated in the virtual pull-back (`Backward.cell_collision`); `shape.volume_weights` $g_c = -\sum_k C_{k,c}/V_c$; `shape.volume_cotangent` differentiates a smoothed-Heaviside flow volume (32^2 samples per cell, half-width $h_\phi/4$, split cells by their region raster). Check: $\sum_c \mathrm{d}V_c/\mathrm{d}s = -7.068$ against $-\pi R^2 = -7.069$; weighted sums match FDs of the forward's MC volumes to their noise.

Finding 3: the along-flight identity drops the collision operator

The identity used is $v \cdot \nabla_x \lambda = -T$; with collisions it is $v \cdot \nabla_x \lambda = -T - \mathcal{L}_f^* \lambda$. The note justifies the omission at $\leq 0.3\,\%$ collisions per particle-step.

squashkn at Kn $= 0.5$ ($\lambda_{\mathrm{mfp}} = 2R \cdot 0.5 = 1.5\,\mathrm{m}$; $\bar{c} \approx 380\,\mathrm{m/s}$):

- M 0.32, $\Delta t = 10^{-4}$: $\nu \Delta t \approx \bar{c} \Delta t / \lambda_{\mathrm{mfp}} \approx 2.5\,\%$ per particle-step, about $8\times$ that bound;
- M 10, $\Delta t = 2.5 \times 10^{-5}$: $\approx 0.6\,\%$ in the free stream, several per cent in the shock layer;
- over a cell crossing ($h = 0.5\,\mathrm{m}$): $h/\lambda_{\mathrm{mfp}} \approx 1/3$ of a collision.

So the along-flight component of $\nabla_x \lambda$ misses an $O(h/\lambda_{\mathrm{mfp}})$ part at Kn $= 0.5$. The pooled one-sided fit (`one_pool`) has no such omission. The four variants run in one scan, so comparing `one_pool` with `along_pool` on the squashkn seeds costs almost nothing extra. The arbiter is the Kn $= 0.5$ FD gate (G4), which has not been run.

Finding 4: hits that are not differentiated

The replay and the backward mask non-survivors. Two kinds of body hits therefore get neither a reward in λ nor a VJP:

- entrants' birth-step hits (inflow tracks them for $\xi \in [0, \Delta t]$);
- hits by a particle that leaves the box in the same step.

Their share is `newborn_hit_share` $= (D_{\text{tally}} - \sum r)/D_{\text{tally}}$. It is recorded, not enforced. It is 0 while the body is more than $|v|\Delta t$ from the inflow and outflow faces. It is nonzero for the `optc` Adam bodies that reached those faces, which are already flagged as not quotable.

Recommendation: fail (or warn) when `newborn_hit_share` exceeds about 10^{-3}.

Left out on purpose: the silhouette term for specular bodies ($O(\varepsilon_b)$); the grazing floors (measured: $+0.8 \pm 0.8\,\%$ at 0.05); the ridge $\rho = 10^{-8}$. The timing term of the drag is *in* since 2026-10-03 (Finding 6).

Bias study (2026-10-02/03): how the runs were made

Small runs on the 4 GB laptop GPU, every estimator paired with the CRN FD of the same seed (`scratch/bias/`: `kn_gate.py`, `fm_exact.py`, `horizon.py`, `summarise.py`).

- Squash ellipse ($R = 1.5$, level set $h_\phi = 0.0625$, grazing 0.05), warm-up 700, 300 captured steps, window 200, CRN FD $h = 0.05$.
- Free-molecular: fnum$\times$2 (75 particles/cell; fnum$\times$1 no longer fits with the new outputs). Kn $= 0.5$: fnum$\times$2, virtual pull-back $S \times K = 2 \times 4$.
- All $\nabla_x \lambda$ variants from one replay (`shape.gradient_variants`), so their differences are paired too.

An exact free-molecular reference that failed. Without collisions each path is a deterministic function of θ, so the forward-mode JVP of the replayed wall map, chained step to step, is the exact pathwise derivative (`fm_exact.py`; it reproduces the forward's drag to 2×10^{-16}). The squash box is a dispersing billiard (convex body, specular y walls): tangents grow at every body bounce, and the chained JVP is heavy-tailed — from step 0 it ranged from $-5.5\times$ to $+8.8\times$ the tally term over 8 seeds, and even over the 300 captured steps from $0.89\times$ to $1.40\times$ (the true $\nabla\lambda$ parts are a few per cent). Useless per seed; the fits' smoothing is what makes the ensemble adjoint usable here.

Free-molecular specular squash: adjoint − FD, % of FD

variant	$s = 0.35$		$s = 0.6$		$s = 1$	
	no timing	+ timing	no timing	+ timing	no timing	+ timing
`one`	-3.0 ± 0.8	-2.1 ± 1.2	$+1.3 \pm 4.1$	$+1.3 \pm 3.9$	-1.8 ± 2.3	-1.7 ± 2.3
`one_pool`	-19.4 ± 0.7	-18.4 ± 1.1	-5.1 ± 1.3	-5.1 ± 1.8	-2.3 ± 1.5	-2.3 ± 1.6
`along`	-1.7 ± 1.2	-0.7 ± 1.7	-0.7 ± 2.4	-0.7 ± 2.4	-0.9 ± 1.5	-0.9 ± 1.5
`along_pool`	-9.9 ± 1.3	-9.0 ± 1.7	-4.5 ± 1.4	-4.5 ± 1.9	-2.1 ± 1.4	-2.1 ± 1.6
`one_pool_plane`	-2.7 ± 0.6	-1.7 ± 0.9	-2.6 ± 1.2	-2.6 ± 1.9	-1.5 ± 1.5	-1.5 ± 1.6
`along_pool_plane`	-1.7 ± 1.1	-0.7 ± 1.5	-2.5 ± 1.4	-2.5 ± 1.9	-1.5 ± 1.4	-1.4 ± 1.5

8 CRN seeds per s (fnum×2), ±1 SE of the paired difference. "+ timing" adds the window-edge term (Finding 6; 0.2-step band at $s = 0.35, 0.6$, 10-step band at $s = 1$).

- `one_pool`, `along_pool` (the cluster default) are $-19\,\%$ / $-9\,\%$ at $s = 0.35$ and $-5\,\%$ at $s = 0.6$: Finding 5.
- Restricted to the hit's side (`*_plane`) each entry is within 2σ; pooled over s a residual $-1.9 \pm 0.7\,\%$ remains (not located: wall re-hits before the captured horizon, the grazing cut-off).
- The hit-ahead class (not in the table), $s = 1$, no timing: $+4\,\%$ (`one`, $z = 2.3$), $+6\,\%$ (`along`, $z = 6.6$), $+20\,\%$ (pooled, $z \approx 10$). Finding 1 holds in free-molecular flow too.
- The grazing cut-off 0.05 changes the truncated JVP by $+0.8 \pm 0.8\,\%$.

Finding 5: the pooled fit reaches across a thin body

`pool = 1` draws the $k = 96$ nearest nodes from the 3×3 block of cells around the hit's cell (1.5×1.5 m). At $s = 0.35$ the body is $2sR = 1.05$ m thick — two cells — so the block around a hit on the upper face holds the particles below the body. Their values have nothing to do with the query's, and the one-sided rule ($v \cdot n > 0$) does not separate them: below the body $v \cdot n_{\text{top}} > 0$ means moving *towards* the lower face.

Fix (`ensemble_device.node_gradients(halfspace=)`): a one-sided query also drops the nodes more than `halfspace` cells behind the plane through the query normal to its hit normal,

$$(x_{\text{node}} - x_q) \cdot n_{\text{hit}} \geq -\tfrac{1}{4}\,\Delta x.$$

Variants `one_pool_plane`, `along_pool_plane` (`shape.VARIANTS`); the latter is now the default of `optimize.py`, `optimize_squash.py` and `ensemble_mesh`. It also removes most of the v part's spread ($-1.1 \to -0.12 \times 10^{-21}$ at $s = 0.35$).

Finding 6: the window-edge timing term

The per-hit drag's window edges make $\mathbb{E}J$ depend on which step a hit falls in. The band estimator (Term 4) uses the hits within a band of each edge. It was left out ("zero mean on a steady window").

- The window is steady: the drag's relative change over the 200 steps is $-0.1\,\%$ (specular) and $+0.8\,\%$ (diffuse), $\pm 2\,\%$ per 20-step bin.
- 0.2-step band (one step per edge, ~ 10 hits per seed): $+2.8 \pm 1.2\,\%$ at $s = 1$ on 8 seeds, and $-35\,\%$ on one diffuse seed. This first looked like the missing $-2.4\,\%$ of the old fnum$\times$1 gate; it is noise.
- 10-step band (`timing_band` ≥ 1: whole steps per edge, `ensemble_mesh.timing_band_setting`), same 8 seeds: $+0.03 \pm 0.25\,\%$.

Decision: the term is part of the windowed objective's derivative (nonzero whenever the window is not steady), so it is now in the reported gradient, with the 10-step band as the default (`optimize.py`, `optimize_squash.py`, `ensemble_mesh`).

Diffuse squash body (acc = 1), free-molecular, $s = 1$

variant	$s = 1$	
	no timing	+ timing
`one`	$+1.0 \pm 3.1$	$+1.2 \pm 3.0$
`one_pool`	-6.0 ± 3.1	-5.8 ± 3.0
`along`	-0.6 ± 3.4	-0.4 ± 3.3
`along_pool`	-3.2 ± 3.4	-3.1 ± 3.3
`one_pool_plane`	-3.1 ± 3.1	-2.9 ± 3.0
`along_pool_plane`	-1.6 ± 3.4	-1.4 ± 3.3

8 CRN seeds, fnum×2, silhouette band 0.03, grazing 0.05; adjoint − FD in % of FD. The first diffuse squash gate: the gradient is

$$\underbrace{9.12}_{\text{tally (hybrid)}} + \underbrace{9.84}_{\text{silhouette}} - \underbrace{0.13}_{\text{normal score}} + \nabla\lambda \text{ parts } (-0.15) \quad [10^{-21}\,\mathrm{N/m}],$$

against FD 19.0: the silhouette term is half of it, and every variant but the cross-body `one_pool` is within its ±3 % noise ("+ timing": the 10-step band, $+0.2 \pm 0.8$ %). The 0.2-step band had a 6 % SE here — a diffuse hit's reward does not vanish at grazing incidence — one seed read −35 % (Finding 6).

Finding 2 resolved: the cut-volume term, built and gated

Machinery gate (`scratch/bias/volume_scale.py`, Kn $= 0.5$, fnum$\times$4, $S \times K = 2 \times 4$, 8 CRN seeds): scale the NTC volume of a set of cells by $1 + \epsilon$ — the only change — and compare the FD in ϵ ($\epsilon = \pm 0.2$) with the adjoint's $\sum_{c \in \text{set}} g_c V_c$:

cells scaled	FD	adjoint	z
all	2.11 ± 0.26	2.45 ± 0.08	-1.2
cut cells	0.69 ± 0.45	0.34 ± 0.01	$+0.8$

(10^{-3} N/m)

Geometry factor: $\sum_c \partial V_c / \partial s$ from the smoothed VJP $= -7.068$ against $-\pi R^2 = -7.069$; random-weight sums match FDs of the forward's own MC volumes.

Size on the squash: $\mathrm{d}D/\mathrm{d}s|_V = -4.4 \times 10^{-4}$ at $s = 1$ (-2.9 % of $\mathrm{d}D/\mathrm{d}s$, seed SE 0.1 %). The squashkn cluster runs lacked it.

Finding 3: the along-flight identity with collisions

Kn $= 0.5$, $s = 1$, fnum×2, $S \times K = 2 \times 4$, 8 seeds. The fitted slope (`*_pool_plane`: no identity) against the identity's value of its component along v, paired per seed (x part, 10^{-4} N/m):

$$\text{fit} - \text{identity}(T + \mathcal{L}^*\lambda) = -1.3 \pm 0.5, \qquad \text{fit} - \text{identity}(T) = +4.9 \pm 0.5.$$

- The identity without $\mathcal{L}^*\lambda$ (the cluster runs) is off by 3.2 % of $\mathrm{d}D/\mathrm{d}s$ ($z \approx 10$); with it the fit and the identity nearly agree.
- The residual -0.8 % ($z = 2.5$) is within what the discrete scheme allows: NTC gives a particle one collision chance per step wherever it sits in its cell, so a shift along v is a position offset, not a delay; the continuous identity holds to $O(\nu \Delta t)$.
- All variants are within ± 1.5 % of each other once $\mathcal{L}^*\lambda$ is in.

variant	$s = 1$	
	no timing	+ timing
`one`	-7.3 ± 6.1	-7.2 ± 6.2
`one_pool_plane`	-8.0 ± 6.0	-7.8 ± 6.1
`along`	-6.2 ± 5.9	-6.0 ± 6.0
`along_free`	-9.9 ± 5.9	-9.8 ± 6.0
`along_pool_plane`	-7.2 ± 6.1	-7.1 ± 6.2
`along_pool_plane_free`	-11.0 ± 6.1	-10.8 ± 6.1

against the paired FD ($h = 0.05$, ~ 18 % per-seed noise);

precise FD: next slide.

Kn = 0.5: the remaining gap and what it is not

Precise FD (`scratch/bias/fd_only.py`, 32 CRN seeds, fnum×2): $h = 0.1$: 0.01650 ± 0.00019, $h = 0.2$: 0.01605 ± 0.00010, Richardson 0.01665 ± 0.00026 (the paired $h = 0.05$ FD has 18 % per-seed noise; $h = 0.2$ has 3.5 %).

Adjoint (8 seeds, with volume and timing terms): `along` -6.9, `along_pool_plane` -7.9, `one_pool_plane` -8.7, old identity -11.6 % (each ± 1.8). Tally + volume alone: 0.0164.

Ruled out (paired, same forward):

- horizon: 300 → 450 captured steps (fnum×3, $S \times K = 2 \times 8$, 3 seeds): $+0.1 \pm 0.4$ % for the fit variants (the identity variants $+3.5 \pm 3.5$ %);
- virtual quadrature: $S \times K = 2 \times 4 \to 4 \times 8$: < 1 % (2 seeds);
- fit width: $k = 48, 96, 192$ pooled: within 1 %; unpooled $k = 16$ reads +4–10 % higher (noisy);
- one-sided backward fits next to the body (`backward_normals`): < 1 %;
- timing band: +0.15 %.

Open: the $\nabla\lambda$ parts (x +2 %, v −10 % of $\mathrm{d}D/\mathrm{d}s$) are the only pieces large enough; the exact (sequential) pull-back on the same seeds decides between the virtual quadrature and the fits / linearisation.

The free-molecular $-2\,\%$: noise or bias?

Per s, adjoint $-$ FD (%), 8 seeds each, fnum$\times$2; pooled with inverse-variance weights:

	$s = 0.35$	0.6	1	pooled
`along_pool_plane`	-1.7	-2.5	-1.5	-1.8 ± 0.7 ($z = -2.5$)
`one_pool_plane`	-2.7	-2.6	-1.5	-2.6 ± 0.5 ($z = -5.3$)
tally term alone	-1.2	-2.6	-1.2	
tally alone, fnum$\times$1 (old gate)	-2.6	-2.4	-2.4	($\pm$0.9–1.4 each)

Systematic (χ^2 across s consistent with one offset; two independent data sets). It sits in the *tally* term, which is exact given the hits — the fits are not the cause.

- FD curvature (central difference, $h = 0.05$, on the analytic reference): $\leq 0.15\,\%$, and of the wrong sign.
- Grazing cut-off: removing it (same forward) moves the gradient $+0.15 \pm 0.25\,\%$.
- **The horizon**: particles reflected *before* the captured steps bounce off the specular y walls ($\sim$230 steps round trip) and re-hit inside the window, and the adjoint cannot see their first hit's dependence on the shape. Capturing 600 instead of 300 steps (same forward, fnum$\times$4, 8 seeds): $+0.8 \pm 0.25\,\%$ at $s = 1$, $+1.1 \pm 0.3\,\%$ at $s = 0.35$. Most of the $-2\,\%$ is truncation, not a code error; the cluster jobs `fm_hor_s*` capture the whole run (no truncation at all) to close it.
- Consequence: the box's specular walls give the free-molecular case a memory of a few hundred steps; a captured horizon ≥ 600 steps (or open y faces) for gates at the 1 % level.

Audit: where could a no-collision assumption hide?

The step is move (hits) $\rightarrow$ NTC collisions. Each term's justification, re-checked with collisions:

term	assumption	with collisions
pathwise x, v	$\nabla\lambda_k$ at the post-move state; λ_k includes step k's collision	holds
tally, score, hybrid normal	exact reward; weights λ + later rewards	holds
silhouette	straight flight to the step end	holds (collisions follow the move)
timing band	a hit changing step matters only at the window edges	**wrong**: the hit/collision order swaps — Finding 7
along-flight	continuous identity $v \cdot \nabla_x \lambda = -T - \mathcal{L}^* \lambda$	$O(\nu \Delta t)$ in the discrete scheme (Finding 3); fit-only variants avoid it
cut-cell volume	—	built (Finding 2)
cell membership of a hit particle	omitted (integer)	$O(\nu \Delta t)$ below the order term
`hit_fits`	only hit slots' post-move states depend on θ	holds

Finding 7: the order of a hit and the step's collision

NTC collides once per step, after the move. A body hit at the end of step k that the shape change pushes into step $k+1$ collides in step k as the *incoming* particle (x_w, v^{in}) instead of the *reflected* one (x_w, v^{out}); both futures coincide after the next move, so the objective changes by

$$J_c = c(x_w, v^{\mathrm{in}}) - c(x_w, v^{\mathrm{out}}), \qquad c = \Delta t\, \mathcal{L}^* \tilde{\lambda},$$

at *every* step boundary (the window-edge reward jump is the collisionless remainder of the same event). Without collisions $c \equiv 0$. Estimator (`shape.order_jump`, `ShapeTerms.order`): c fitted from the backward's per-slot increments on the approaching and leaving sides at the hit point, whole-step band (hit times uniform within a step), cotangent $J_c/\Delta t$ on every body hit's time.

Kn $= 0.5$, $s = 1$, 8 seeds: $+6.3 \pm 4.4 \times 10^{-5}$, i.e. $+0.4 \pm 0.3\,\%$ of $\mathrm{d}D/\mathrm{d}s$; the gap stays -6.5 to $-8.3\,\%$ (`along` to `one_pool_plane`). Small, because the incoming/reflected collision effects differ by a small fraction of the reward; in the gradient by default.

Finding 1, final: the hit-ahead class cannot be repaired

The class κ = "a body hit at a later step" is a property of the particle's *future*. With $\lambda = P\lambda_1 + (1-P)\lambda_0$, a fit inside the class estimates $P\nabla\lambda_1 + (1-P)\nabla\lambda_0$ and drops $\nabla P(\lambda_1 - \lambda_0)$. Restoring it needs ∇P from the same local regression of the class indicator — and that sum is the mixed-pool fit again, with more variance. It is unbiased only where P is constant over the fit's support, which no case here satisfies (re-hits off the box walls; collisions; diffuse re-emission). Measured: $+40\,\%$ ($\cos 2\theta$), $+4$–$20\,\%$ (free-molecular squash).

What survives of the idea: stratify by a function of the *current state*. The one-sided $v \cdot n$ rule and the tangent-plane half-space (Finding 5) do exactly that, and are unbiased wherever λ is smooth inside each piece.

`value_class="hit_ahead"` is kept, documented as diagnostic only, and warns when used.

Validation status (CRN finite differences)

case	result	note
level-set circle radius	$z = -1.4$; = analytic body to 10^{-4} per seed	free-mol.
sphere radius (8 seeds)	$z = -1.71$	free-mol.
specular capsule width, hit-ahead fit	1.356 ± 0.011 vs FD 1.35, $z = +0.08$	free-mol.
diffuse plate (silhouette + hybrid normal)	1.191 ± 0.068 vs 1.17–1.34, $\lvert z\rvert \leq 1.6$	free-mol.
$\cos 2\theta$, coarse, before the class	$z = +0.7, -0.2$	free-mol.
$\cos 2\theta$, fine, with the class	-2.386 ± 0.005 vs -1.59, $z \approx -120$	Finding 1
$\cos 2\theta$, coarse, no class (4 seeds)	paired gradient $-$ FD $= +0.007 \pm 0.071$	free-mol.
lane vs stencil tracker	agree to 2.3×10^{-15}	
squash, specular, $s = 0.35/0.6/1$	$-0.7/-2.6/+0.4\,\%$ (± 2)	free-mol.
squash $s = 1$, diffuse	$-1.4 \pm 3.3\,\%$ (10-step timing band)	free-mol.
cut-volume machinery (volume scaling)	$z = -1.2$ (all), $+0.8$ (cut)	Kn 0.5
squash $s = 1$, Kn $= 0.5$	-7 to $-9 \pm 1.8\,\%$	**open**

Free-molecular shape gradients, specular and diffuse, are FD-consistent to $\sim 2\,\%$ with the new defaults. The collisional gradient is 7–9 % low after the volume and $\mathcal{L}^*\lambda$ terms; the cause is not yet located (previous slide).

Status after the bias study, and next steps

Changed on main (uncommitted)

- `value_class="none"` default (Finding 1); `along_pool_plane` default variant (Finding 5); window-edge timing term in the reported gradient (Finding 6);
- cut-volume term in every collisional gradient (Finding 2); $\mathcal{L}^*\lambda$ in the along-flight identity (Finding 3); warning above 10^{-3} newborn-hit share (Finding 4);
- `backward_virtual(collision_out=, columns_out=)`; `backward_normals` option.

Next

1. Locate the Kn $= 0.5$ gap: cluster-scale gate at fnum$\times$1, $S \times K = 4 \times 8$ with the FD-only arbiter at $h = 0.1, 0.2$; a velocity-local fit test (the unpooled $k = 16$ fit read higher); the small-$v \cdot n$ structure of λ for reflected particles.
2. Re-run the squashkn optimisations only after that gap is closed; the free-molecular squash and `optc` studies can be re-run now with the new defaults (the old runs used `along_pool`: $-9\,\%$ at $s = 0.35$, and `optc` the class: $+40\,\%$).
3. Use a timing band of whole steps (`timing_band` ≥ 1) for diffuse bodies.